

Parallel and Distributed Computing (CS3006)

Date: February 24th, 2025

Sessional-I Exam

Total Time (Hrs.): 1

Total Marks: 60

Total Questions: 6

Course Instructor(s)

Dr. Abdul Qadeer

Mr. Danyal Farhat

Mr. Raja Muzammil Muneer

Ms. Saba Tariq

Dr. Syed Muhammad Irteza

Dr. Zeeshan Ali Khan

Roll No

Section

Student Signature

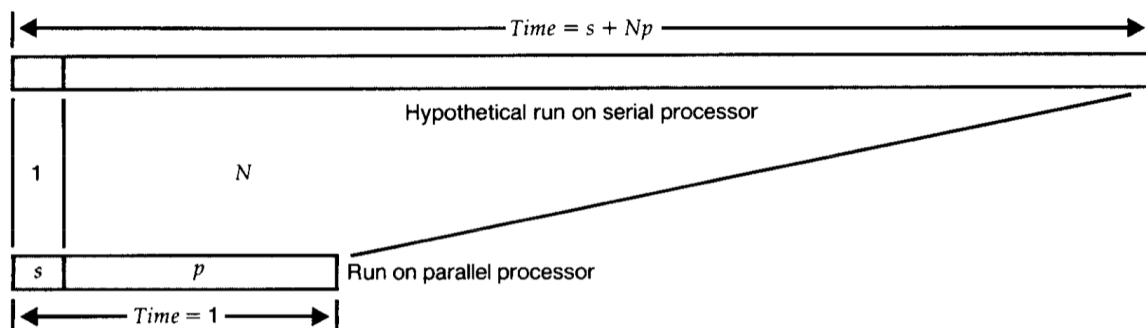
Instructions / notes:

1. Attempt all questions on the answer sheet.
2. If a question is ambiguous, note it, assume clarifying details, and solve it.
3. Include all work with your answer; do not use a separate rough sheet.
4. We estimate that you need no more than 10 minutes on any question. Pace yourself accordingly.

CLO 3: Perform analytical modelling, dependence, and performance analysis of parallel algorithms and programs.

Q1a: [2 marks] How does Gustafson's law differ from Amdahl's law?

Q1b: [4 marks] Derive Gustafson's law using the symbols provided in the diagram below. Clearly mention what each symbol means. Your final expression should be in terms of N and s only.



Q1c: [2 marks] What are the two assumptions made in the above derivation?

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Solution for Q1:

Q2a: In Gustafson's Law we increase the input data with increasing processors. In Amdahl's law we keep the input data same across runs with increasing processors.

Q2b: We are increasing data that can be used in parallelization in proportion to the number of processors (N). Assuming a single processor will take N times more time to finish the work, the time of a hypothetical serial run = s (serial proportion of time) + N*p
We assume that using N processors the time will be: s + p = 1

Scaled speed-up is as follows:

$$\begin{aligned} \text{Scaled speedup} &= (s + p \times N) / (s + p) \\ &= s + p \times N \quad \text{Because } s+p = 1 \\ &= N + (1 - N) \times s. \quad \text{Put } p = (1-s) \text{ in above equation and simplify.} \end{aligned}$$

This is a line having slope (1-N)

s = serial fraction

p = parallel fraction

N = number of processors

Q2c: We assumed that each processor is equally capable and can divide and finish its allocated work at the same rate.

We assumed there are no overheads and data-based scaling will perfectly continue with increasing processors.

CLO 3: Perform analytical modelling, dependence, and performance analysis of parallel algorithms and programs.

Q2a: [2 marks] Write Amdahl's law and Karp-Flat metric (K-F metric) formula (no derivation needed).

Q2b: [4 marks] Assume that 90% of the algorithm can be parallelized while the remaining 10% must be executed sequentially. We have partially filled the following table for you with 4 missing values (A to D). Calculate these four missing values using formulae you mentioned above.

Number of Processors used	2	4	6	8
Amdahl's speedup	A	3.077	4	4.706
Gustafson's speedup	B	3.7	5.5	7.3
K-F metric using Amdahl's speedup	C	0.1	0.1	0.1
K-F metric using Gustafson's speedup	D	0.027	0.018	0.014

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Q2c: [2 marks] Review the K-F metric derived from Amdahl's speedup in your completed table. What conclusions can you draw, and why?

Q2d: [2 marks] Review the K-F metric derived from Gustafson's speedup in your completed table. What conclusions can you draw, and why?

Q2e: [2 marks] Above we calculated K-F metric values using speed-up values provided by Amdahl's or Gustafson's instead of getting the speed-up values by empirically running the code. If we had experimentally computed K-F metric values what trend of K-F values will you expect and why (only for Amdahl's case)?

Solution for Q2:

Q2a: Amdahl's law: $\text{Speed-up} = 1 / (F + (1-F)/N)$, where F = serial fraction, N = no of processors
K-P Metric = $f = (N/\text{speed-up} - 1) / (N - 1)$, N = number of processors

Q2b:

A = 1.818

B = 1.9

C = 0.1

D = 0.053

(Q2c) K-F metric using Amdahl's speedup: serial fraction remains constant as we increase the number of processors as the problem size is fixed. Because we used speed-up calculated via Amdahl's (and not the empirical speed-up observed), K-F is not able to show any overheads.

(Q2d) K-F metric using Gustafson's speedup: serial fraction decreases with scaled speedup since parallelizable workload increases in fraction.

Q2e: Experimental speedups consider overheads i.e. data parallelization overheads which theoretical speedup do not. So we expect that K-F values will increase.

CLO 3: Perform analytical modelling, dependence, and performance analysis of parallel algorithms and programs.

Q3a: [2 marks] You are designing a cloud-based image processing service where each uploaded image is independently transformed (e.g., resized and enhanced). With thousands of concurrent uploads, which Flynn's Taxonomy architecture best supports parallelism and scalability? (Just name it.) Provide rationale for your answer in a line.

Q3b: [8 marks] You're optimizing a scientific computation program and profiling reveals that 70% of the work is parallelizable while 30% remains sequential. You plan to rent a server from a cloud

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provider that offers machines ranging from 2 to 500 cores. While you might be tempted to get a server with 500 cores assuming each extra core will give you some speed-up, but you want an informed decision. You only like speed-ups in whole numbers (such as 5, 6, etc. and not fractions like 5.3 etc.). What core count should you get?

Solution for Q3:

Q3a) Multiple answers are possible:

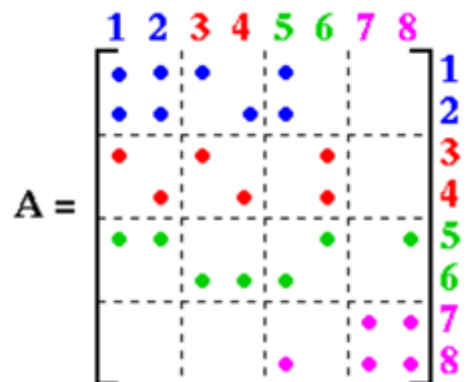
MIMD: Use multiple independent processors working on independent pictures for best performance and scalability

SPMD or SIMD: Since Flynn's taxonomy is quite loose, students can say that it resembles SIMD because all processing elements are doing the same operation but on independent data.

Q3b): We know from Amdahl's that the max speed-up can be $1/F$ even if we use infinite processors. So that number turns out to be 3.33333... Now as per question, we want to target speed up of 3. That means by using that value we get answer of 21 cores.

CLO 3: Perform analytical modelling, dependence, and performance analysis of parallel algorithms and programs.

Q4a: [6 marks] Consider the problem of computing the product $y = Ab$, where A is a sparse 8×8 matrix and b is a dense 8×1 vector. A matrix is considered sparse when a significant number of its entries are zero, and the locations of the non-zero entries do not conform to a predefined structure or pattern. The matrix A is shown on the right, with the non-zero elements represented by dots. Create a task interaction graph for this problem, such that i^{th} task computes entry of $y[i]$ and owns the i^{th} row of A and the element $b[i]$.



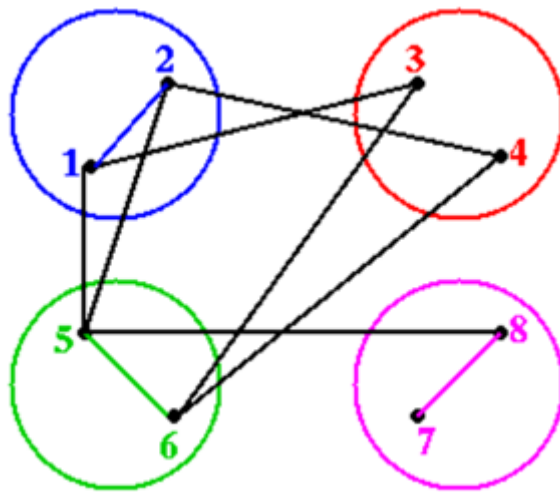
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Q4b: [4 marks] Map the tasks created in part (a) to four processes, such that two consecutive tasks are assigned to one process. Calculate the list C_i , where $i = 1$ to 4, which contains the indices of b that the tasks on Process i need to access from tasks mapped to other processes.

Solution of Q4:

Solution for Q1, parts (a) and (b)



C1 (blue colored) = (3,4,5)

C2 (red colored) = (1,2,6)

C3 (green colored) = (1,2,3,4,8)

C4 (purple colored) = (5)

CLO 1: Demonstrate understanding of various concepts involved in parallel and distributed computer architectures.

Q5 a: [2 marks] Draw a 4x4 Omega network for processors P_0 – P_3 and memory modules M_0 – M_3 .

Q5 b: [2 marks] Explain how P_3 will access M_2 in terms of routing?

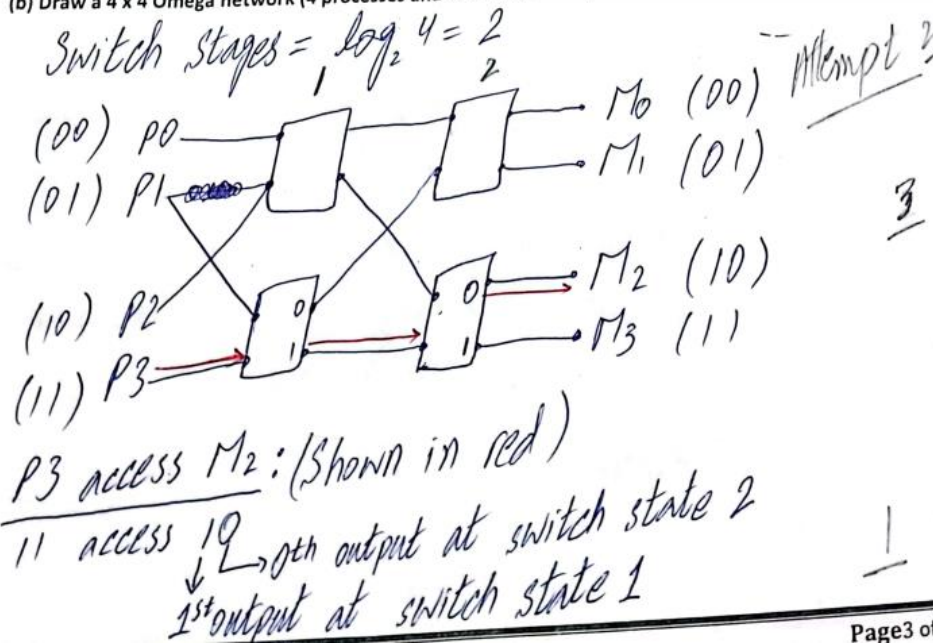
Q5 c: [2 marks] How many switching stages will be required for this network?

Q5 d: [2 marks] Can P_1 to M_2 and P_3 to M_3 communications happen concurrently? Why or why not?

Q5 e: [2 marks] How many switching nodes will be required for this network?

Solution of Q5:

(b) Draw a 4 x 4 Omega network (4 processes and 4 memory modules)



(c) Switching Stages = 2

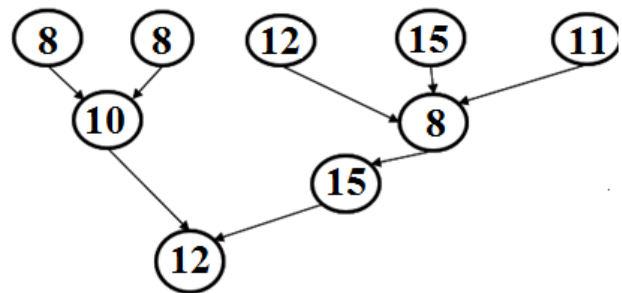
(d) Not possible. There is a conflicting link on the path between bottom two switches.

(e) Switching Nodes = $(p / 2) \times \log(p) = (4/2) \times (\log(4)) = 2 \times 2 = 4$

CLO 3: Perform analytical modelling, dependence, and performance analysis of parallel algorithms and programs.

Q6: [2+2+2+2+2 marks] Examine the task-dependence graph on the right and compute:

- Maximum degree of concurrency
- Any Single Critical Path Work
- Average degree of concurrency
- Minimum number of processors needed to obtain the maximum possible speedup
- Maximum speed up if number of processors are limited to 4



Solution of Q6:

- Maximum degree of concurrency
Maximum degree of concurrency = 5
- Any Single Critical Path Length
Critical Path Length = $15 + 8 + 15 + 12 = 50$

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(iii) Average degree of concurrency

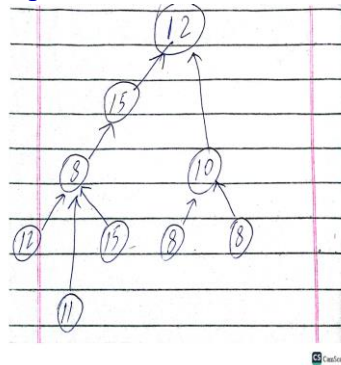
Average degree of concurrency = $(8 + 8 + 12 + 15 + 11 + 10 + 8 + 15 + 12)/50 = 99/50 = 1.98$

(iv) Minimum number of processors needed to obtain the maximum possible speedup

Minimum number of processors needed to obtain the maximum possible speedup = 5

(v) Maximum speedup if number of processors are limited to 4

Maximum speedup if number of processors are limited to 4 = $9 / 5 = 1.80$



Parallel and Distributed Computing (CS3006)

Date: Sep 24th 2024

Course Instructors:

Mr. Danyal Farhat

Dr. Zeeshan Ali Khan

Sessional-I Exam

Total Time: 1 Hour

Total Marks: 30

Total Questions: 5

Semester: Fall-2024

Campus: Lahore

Dept: CS

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CLO # 1: Demonstrate understanding of various concepts involved in parallel and distributed computer architectures.

Q1: Assume a sequential program S has an execution time of 500 seconds. Further, assume that S_p is the execution time of the parallel variant of S . After an experimental evaluation over different number of processors, the following running times were achieved: [2 + 2 + 1 marks]

P	2	4	6	8
S_p Parallel Execution Time (seconds)	294	203.84	176.62	164.92

- Calculate speedups for each of the experimental configurations.
- Calculate the values for the Karp-Flatt metric. Furthermore, also interpret the results of Karp-Flatt metric and write your opinion accordingly.

Q1 (a)

$$500/294 = 1.7006$$

$$500/203.84 = 2.4529$$

$$500/176.62 = 2.8309$$

$$500/164.92 = 3.0317$$

Q1(b) Calculate the values for the Karp-Flatt metric. Furthermore, also interpret the results of Karp-Flatt metric and write your opinion accordingly.

$$e = \frac{1/s - 1/p}{1 - 1/p}$$

$$((1/1.70) - (1/2)) / (1 - 1/2) = 0.18$$

$$((1/2.45) - (1/4)) / (1 - 1/4) = 0.21$$

$$((1/2.83) - (1/6)) / (1 - 1/6) = 0.22$$

$$((1/3.03) - (1/8)) / (1 - 1/8) = 0.23$$

Since 'e' is increasing with p, it suggests that parallelization overhead is also contributing to the poor speedup. Hence, we need to reduce this overhead to improve speedups.

Q2(a)

$$\text{Diameter} = \log_2(p) = \log_2(16) = 4$$

$$\text{Bisection width} = p/2 = 16/2 = 8$$

$$\text{Arc connectivity} = \log_2(p) = \log_2(16) = 4$$

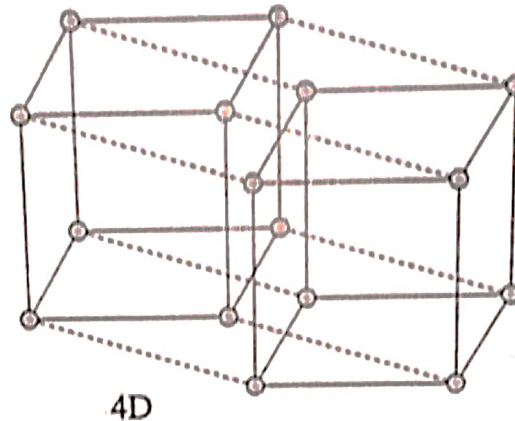
$$\text{Cost} = (p * \log_2(p)) / 2 = (16 * \log_2(16)) / 2 = 32$$

Q2(b)

CLO # 1: Demonstrate understanding of various concepts involved in parallel and distributed computer architectures.

Q2: Calculate diameter, arc connectivity, cost and bisection width for the following architectures:
[2 + 2 marks]

a) Four-dimensional hypercube (size=16 Nodes)



b) Complete binary tree having 3 levels as shown in figure below

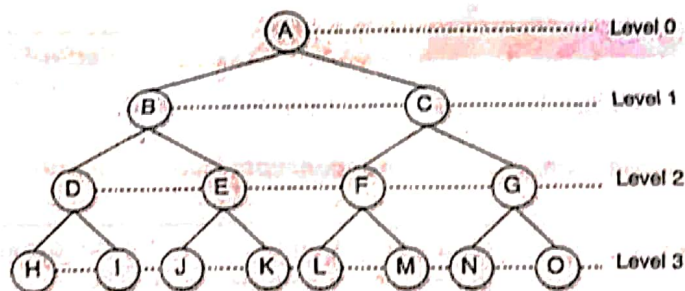


Fig. Complete Binary Tree

CLO # 3: Perform analytical modelling, dependence, and performance analysis of parallel algorithms and programs.

Q3: For the task dependency graph given in figure below, find the following parameters:

- Maximum degree of concurrency
- Total Amount of Work
- Critical Path Length
- Average degree of concurrency

$$\text{Diameter} = 2 \cdot \log((p+1)/2) = 2 \cdot \log(16/2) = 2 \cdot \log(8) = 2 \times 3 = 6$$

$$\text{Arc Connectivity} = 1$$

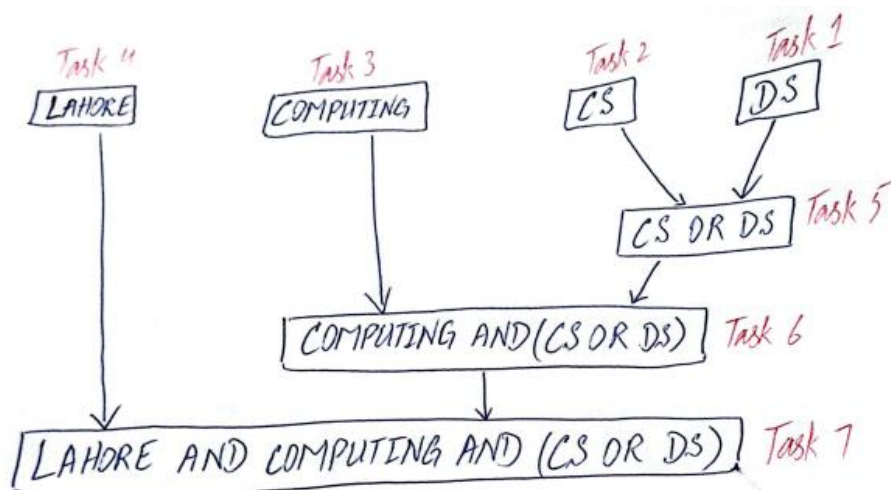
$$\text{Bisection Width} = 1$$

$$\text{Cost} = p - 1 = 15 - 1 = 14$$

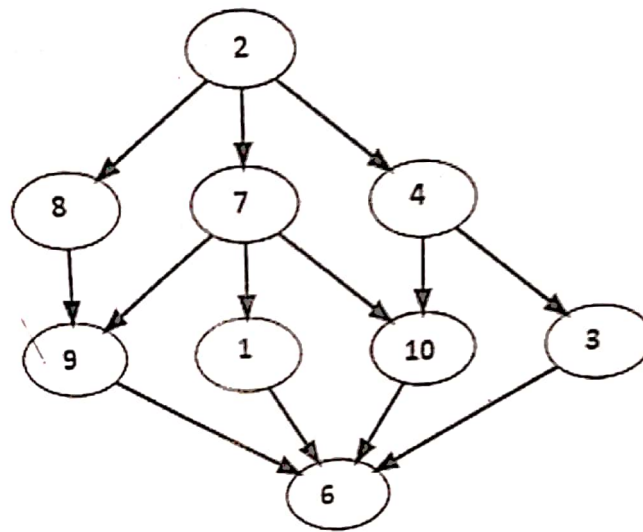
- Q3: (a) Maximum degree of concurrency = 4
 (b) Total Amount of Work = 50
 (c) Critical Path Length = 25
 (d) Average degree of concurrency = 2
 (e)

Tasks having corresponding weights 2, 8, 9, 6 may be mapped to Pa
 Tasks having corresponding weights 7, 1 may be mapped to Pb
 Tasks having corresponding weights 3, 4, 10 may be mapped to Pc

Q4.



7 tasks will be performed for the complete execution of the query.



- e) Furthermore, map the task dependency graph on three processes, namely Pa, Pb and Pc, to reduce the interaction overhead.

[1 + 1 + 2 + 2 + 2 marks]

CLO # 3: Perform analytical modelling, dependence, and performance analysis of parallel algorithms and programs.

Q4: From Flex, we want to enlist the students enrolled in the CS or DS program of FAST School of Computing in the Lahore campus of NUCES. For the query given below, make a simple task dependency graph. Also confirm how many tasks will be performed for the complete execution of the query.

[5 marks]

CAMPUS = "LAHORE" AND SCHOOL = "COMPUTING" AND (PROGRAM = "CS" OR PROGRAM = "DS")

CLO # 2: Implement different parallel and distributed programming paradigms and algorithms using Message-Passing Interface (MPI) and OpenMP.

Q5 [For sections BCS-7A and BCS-7B ONLY]: Consider the following integral:

$$\int_0^1 \frac{1}{1+x+x^2+x^4}$$

- a) Write a C code to compute this integral.
 b) Convert the code written in part (a) into a parallel OPENMP code without using the "omp for" construct.

[3 + 5 marks]

Q5 (a)

```
static long num_steps = 100000;

double step;

void main ()

{
    int i; double x, total, sum = 0.0;
    step = 1.0/(double) num_steps;
    for (i = 0; i < num_steps; i++){
        x = (i + 0.5) * step;
        sum = sum + 1.0/(1.0 + x + x * x + x * x * x * x * x);
    }
    total = step * sum;
}
```

Q5 (b)

```
static long num_steps = 100000;

double step;

#define NUM_THREADS 4

int main(int argc, _TCHAR* argv[]) {
    int i, nthreads;

    double total, sum[NUM_THREADS];

    step = 1.0/(double) num_steps;

    omp_set_num_threads(NUM_THREADS);

#pragma omp parallel
    {
        int i, id, nthrds;

        double x;

        id = omp_get_thread_num();
        nthrds = omp_get_num_threads();

        if(id == 0)
            nthreads = nthrds;

        for (i=id, sum[id]=0.0; i< num_steps; i=i+nthrds){
            x = (i+0.5)*step;

            sum[id] += 1.0/(1.0 + x + x * x + x * x * x * x);
        }
    }

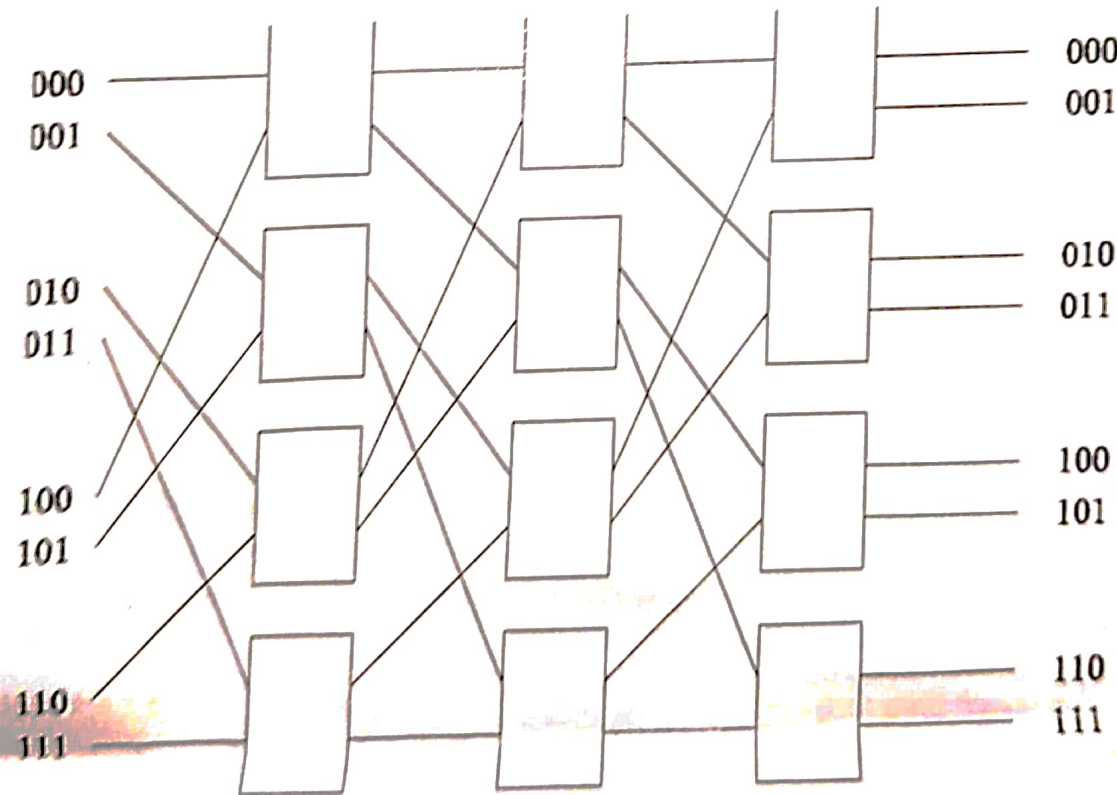
    for(i=0, pi=0.0; i < nthreads; i++)
        total += sum[i] * step;

    printf("%f\n", total);

    return 0;
}
```

CLO # 3: Perform analytical modelling, dependence, and performance analysis of parallel algorithms and programs.

Q5 [For section BCS-7C ONLY]: Draw an 8 x 8 Omega network (8 processes and 8 mem modules) and explain how P3 will access M1. What will be the cost of this network? How many switching nodes will be required for this network?
[2 + 2 + 2 + 2 marks]



Parallel and Distributed Computing (CS3006)

Date: Feb 27th 2024

Course Instructors:

Dr. Abdul Qadeer

Dr. Rana Asif Rehman

Dr. Zeeshan Ali Khan

Mr. Danyal Farhat

Ms. Namra Absar

Sessional-I Exam

Total Time: 1 Hours

Total Marks: 35

Total Questions: 05

Semester: SP-2024

Campus: Lahore

Dept: Computer Science

Student Name

Roll No

Section

Student Signature

Vetted by

Vetter Signature

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- Instruction/Notes**
- Attempt all questions on the provided space in the question paper.
 - Even if you do use rough sheets, they should NOT be attached with final paper.
 - If you find any ambiguity in a question, you can make your own assumption and answer the question accordingly by stating your assumption.
-

Question #	1	2	3	4	5	
Total Marks	8	5	6	8	8	35
Obtained Marks						
CLO #	1	1	1	3	1/3	

CLO # 1: Learn about parallel and distributed computers

Q1: Each MCQ is of 6 points. One or more choices can be right. Select more than one for such cases. [8 marks]

i. How we can classify the type of computer architecture on a typical, consumer-grade multi-core laptop computer released in the year 2024 out of the following?

- A) SISD
- B) SIMD
- C) MIMD
- D) MISD

Correct answer: (A), (B), and (C) because if we just use a single core it will be SISD, if we use all cores concurrently, it will be MIMD. And all modern computers have vector instructions, hence SIMD.

ii. Karp-Flatt metric can give us information for which of the following?

- A) Serial code in terms of number of lines-of-code
- B) Time proportion spent executing serial parts of the code
- C) Any parallelization overheads
- D) Number of processors used for parallelization

Correct answer: (B) and (C)

iii. A network with high connectivity is desirable, because it:

- A) lowers the contention for communication resources.
- B) decreases the energy consumption.
- C) increases the contention for communication resources.
- D) All of the above

Correct answer: (A)

iv. Interpretation of Moore's law assert that:

- A) Computing power will eventually level off
- B) Computing power doubles every 18 months
- C) Transistors decrease in size 50% every 2 years
- D) Data storage costs decrease by 50% every 18 months

Correct answer: (B)

v. Different computers possess the same version of operating system when connected in the form of:

- A) Cluster
- B) Cloud
- C) Network of workstations
- D) None of the given option

Correct answer: (A)

vi. In Bus network, the number of shared connections is equal to:

- A) 1
- B) n
- C) $\log_2 n$
- D) n^2

Correct answer: (B)

vii. Which one is true about Amdahl's law formulation:

- A) It does not account for communication overheads
- B) It provides an upper-bound on the achievable speedups
- C) If the number of processors approaches infinity, Speedup is bounded by $\frac{1}{\text{sequential fraction}}$
- D) All (a), (b), and (c) are correct
- E) Only (b) and (c) are correct

Correct answer: (D)

viii. Choose correct statement/s regarding NUMA architecture:

- A) Access time may vary depending on the data location
- B) Different processors interact with each other using message passing
- C) The same logical address on different processors maps to the same physical memory location
- D) The same logical address on different processors maps to the different physical memory locations

Correct answer: (A) and (C)

CLO # 1: Learn about parallel and distributed computers

Q2: Consider the following equation in the context of Karp-Flatt metric.

[1+1+1+2 marks]

$$f = \frac{1/s - 1/p}{1 - 1/p}$$

(a) If p is the number of processors in the above equation, what is s?

s is speed-up achieved.

(b) What is f in the above equation?

f is the Karp-Flatt metric.

(c) What are the units of f?

It's a unitless entity being a time fraction.

(d) Following data was gathered experimentally on some parallel computer A.

p	Time (seconds)	s	f
1	2.17	---	---
2	1.11	1.95	0.024
3	0.754	2.88	0.021
4	0.577	3.76	0.021
8	0.312	6.96	0.021

What can we infer about the quality of parallelism by looking at the data provided? Please point to specific data to support your answer.

- (1) We are getting speed-up with increasing number of processing elements.
- (2) Karp-Flatt metric is almost the same across runs with increasing number of processing elements. That implies that there might not be any overheads for parallelization though for further improvement we need to target the serial proportion. There is limited amount of parallelism available in the problem.

CLO # 1: Learn about parallel and distributed computers

Q3: Derive generalized expressions for arc-connectivity, diameter, bisection-width, and link cost of a 2D mesh (without wraparound) having M rows and N columns. **[1.5+1.5+1.5+1.5 marks]**

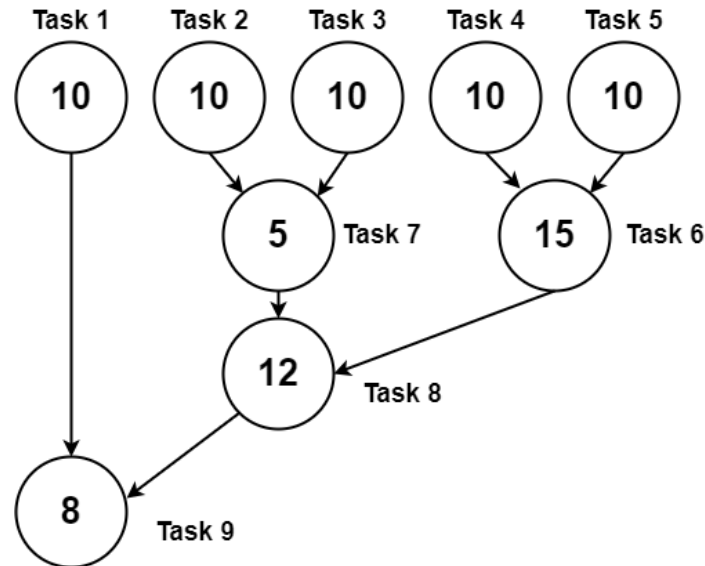
Solution:

$$\begin{aligned}
 (i) \quad & \text{Arc-Connectivity} = 2 \\
 (ii) \quad & \text{Diameter} = \text{Row distance [diameter]} + \text{Column distance [diameter]} \\
 & = (N-1) + (M-1) \Rightarrow M+N-2 \\
 (iii) \quad & \text{Bisection width} = \text{Min of } (M, N) \\
 & \quad \rightarrow \text{Minimum number of links that must be removed to partition in equal halves} \\
 (iv) \quad & \text{Cost} = (\text{Links in a row})(\text{Total rows}) + (\text{Links in a column})(\text{Total columns}) \\
 & = (N-1)(M) + (M-1)(N) \\
 & = MN - M + MN - N \\
 & = 2MN - (M+N)
 \end{aligned}$$

CLO # 3 - Analytical modeling and performance evaluation of parallel programs

Q4: Consider the following task-dependency graph:

[6+2 marks]



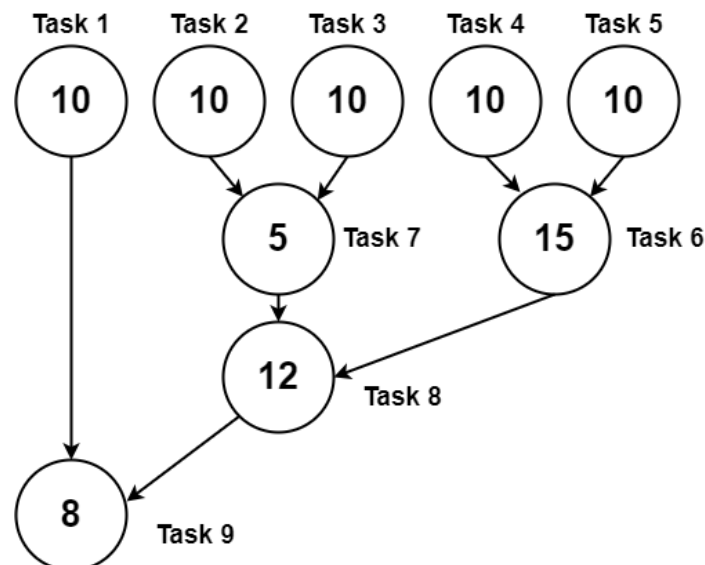
- (a) What is the Maximum degree of concurrency, Critical Path Length, and Average Degree of Concurrency of the task graph?

Maximum degree of concurrency = 5

Critical Path Length = $10 + 15 + 12 + 8 = 45$

Average degree of concurrency = $(10 + 10 + 10 + 10 + 10 + 5 + 15 + 12 + 8)/45 = 90/45 = 2.0$

- (b) Map the task dependency graph on five processes namely, Pa, Pb, Pc, Pd and Pe, to reduce the idling time for the task processing. Please answer using the graph provided below and add a short explanation.



One of the possible solutions is as follows:

Map Tasks 1 and 9 on Pa.

Map Tasks 2 and 8 on Pb.

Map Tasks 3 and 7 on Pc.

Map Tasks 4 and 6 on Pd.

Map Task 5 on Pe.

The mapping will be able to reduce the idling time in the task processing using the maximum degree of concurrency of the task graph.

CLO # 3 - Analytical modeling and performance evaluation of parallel programs

Q5 [For all sections except section BDS-6B]: Recall how merge sort work. We have provided its pseudocode to jog your memory.

[2+2+2+2 marks]

MERGE-SORT(A, p, r)

1 **if** $p < r$

2 $q = \lfloor (p + r) / 2 \rfloor$

3 **MERGE-SORT**(A, p, q)

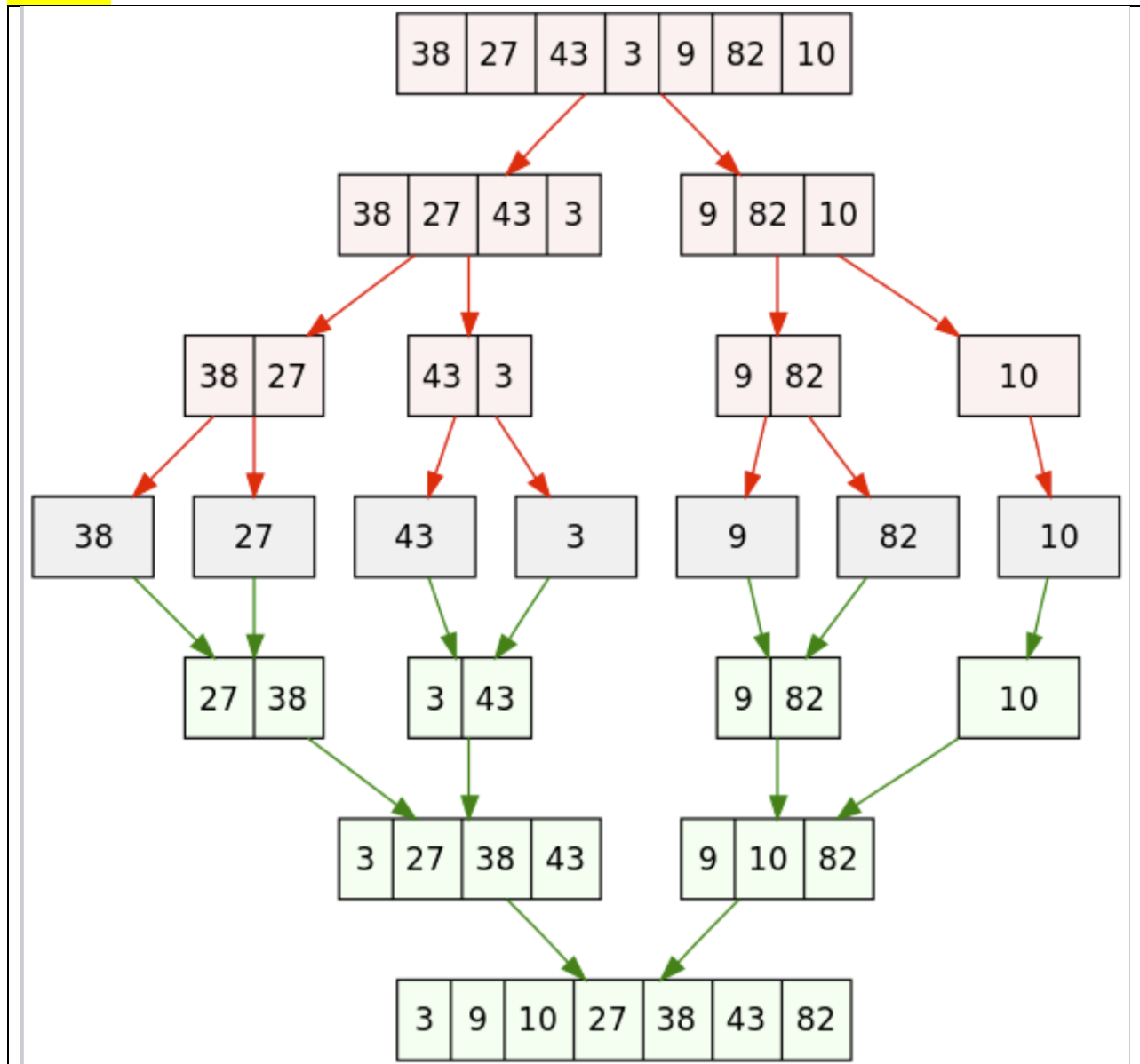
4 **MERGE-SORT**($A, q + 1, r$)

5 **MERGE**(A, p, q, r)

(a) Using the following example data provide a recursive decomposition.

	38	27	43	3	9	82	10	
--	----	----	----	---	---	----	----	--

Solution:



(b) Using the same data as above provide a data-based decomposition.

Divide the data based on the input and equally divide reasonable (but almost equal sized) data to each task. In this example, data is small we can just use two tasks. One will do [38,27,43,3], and the other will do [9,82,10]. They will independently run merge sort and will finally combine the result.

(c) In terms of potential for better concurrency (and better speedup) which one of the above two are a better decomposition approach? (Just name it.)

Data-based decomposition and then combine recursively.

(d) Provide rationale for your above answer logically and succinctly.

For our small example, there seems to be a lot of interactions among tasks in the case of recursive decomposition as the merging starts (for example to bring data from another processor and to merge it, which is a serial work). On the contrary, the data based decomposition initially divides data for processors and then they don't have to interact until they have sorted their part. Since data size is almost same, they will finish about the same time. We could use parallel tasks for merging (for example if we had 4 sorted pieces, we could use 2 processes to merge, and finally one processor to merge).

CLO # 1: Learn about parallel and distributed computers

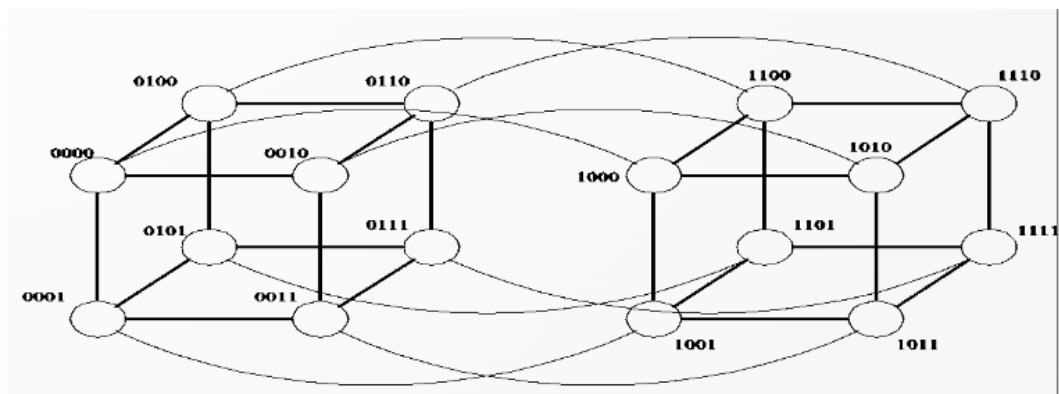
Q5 [For section BDS-6B ONLY]:

(a) Derive the total communication cost for a message of size w words to traverse d communication links using cut-through routing. (Describe each parameter involved.)

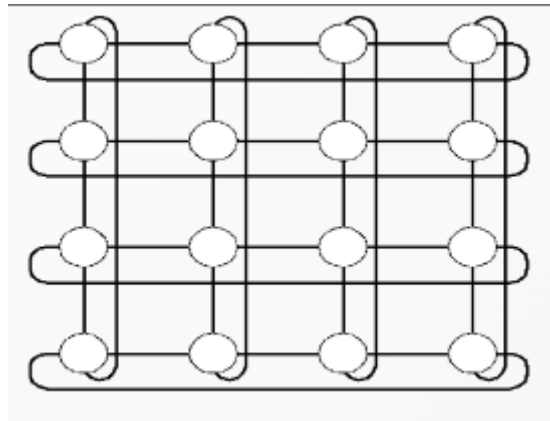
[4 marks]

$$T_{\text{comm}} = T_s + wT_{\text{word}} + dT_h$$

(b) Consider the following two graphs



Graph 1



Graph 2

Compute and compare the diameter and arc connectivity of the given graphs. What does it tell about their performance? **[2+2 marks]**

Diameter of graph1 = $\log p = \log 8 = 3$

Diameter of graph 2 = $2(\sqrt{p}/2) = 4$

arc connectivity of graph 1 = $\log p = 3$

arc connectivity of graph 2 = 4

Graph 1 is more prone to link failure. In case of any failure, there are fewer other routes to the node.

National University of Computer and Emerging Sciences, Lahore Campus

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Student : Name: _____ **Roll No.** _____ **Section:** _____

Instruction: Attempt all questions on the question paper. Rough sheets can be used but it should not be attached. If you think some information is missing then assume it and mention it clearly.

Question # 1: [5+2 marks, CLO # 1]

From the given options, select the best answer.

- i. GPU is an example of
 - a. *Shared Memory SIMD*
 - b. Distributed Memory SIMD
 - c. Shared Memory MIMD
 - d. Distributed Memory MIMD
- ii. Multi-stage Omega network has _____ switching nodes.
 - a. $\log p$
 - b. $p/2 \times \log p$
 - c. $p \times \log p$
 - d. p^2
- iii. The _____ of a task-_____ graph is usually a _____ of the _____ of the task-dependency graph:
 - a. edge-set; interaction; superset; nodes
 - b. edge-set; mapping; subset; edge-set
 - c. edge-set; interaction; subset; edge-set
 - d. *edge-set; interaction; superset; edge-set*
- iv. The use of _____ decomposition comes into play when we _____
 - a. speculative; are not sure of the dependencies between tasks
 - b. speculative; are not sure of the size of the search space from before
 - c. *exploratory; are not sure of the size of the search space from before*
 - d. recursive; want to avoid divide and conquer strategies
- v. We can increase performance of a uniprocessor by
 - a. Increasing clock frequency
 - b. Increasing instructions per cycle
 - c. Multithreading
 - d. *Any one from above can be used*
- vi. 70% of a program's execution time occurs inside a loop that can be executed in parallel and rest 30% in serial. What is the maximum speedup we should expect from a parallel version of the program executing on 8 CPU?
 - a. *2.58*
 - b. 3.06

- c. 4.02
- d. 4.08

Question # 2: [08 marks, CLO # 1]

Assume a sequential program S has an execution time of 500 seconds. Further, assume that S_p is a parallel variant of S . After an experimental evaluation over different number of processors, the following running times were achieved: -

P	2	4	6	8
Execution Time (seconds)	294	203.84	176.62	164.92
Speedup				
Karp-Flatt Metric				

- a) Calculate Speedups for each of the experimental configurations in the space provided below and then write your answers in the table above.

$$500/294 = 1.7006$$

$$500/203.84 = 2.4529$$

$$500/176.62 = 2.8309$$

$$500/164.92 = 3.0317$$

- b) Calculate the Karp-Flatt metric values in the space provided below and then write your answers in the table above. You also have to interpret the results of Karp-Flatt metric and write your opinion accordingly.

$$e = \frac{1/s - 1/p}{1 - 1/p}$$

$$((1/1.70) - (1/2)) / (1 - 1/2) = 0.18$$

$$((1/2.45) - (1/4)) / (1 - 1/4) = 0.21$$

$$((1/2.83) - (1/6)) / (1 - 1/6) = 0.22$$

$$((1/3.03) - (1/8)) / (1 - 1/8) = 0.23$$

Since 'e' is increasing with p, it suggests that parallelization overhead is also contributing to the poor speedup. Hence, we need to reduce this overhead to improve speedups.

Question # 3: [10 marks, CLO # 1]

What is Fat Tree Routing? Draw a Fat tree having 4 levels with 8 leaf nodes. Provide a comparative analysis with a star topology with same number of nodes in terms of cost, diameter, bisection width and arc connectivity. You have to mention which routing is better in terms of each evaluation parameter.

Answer

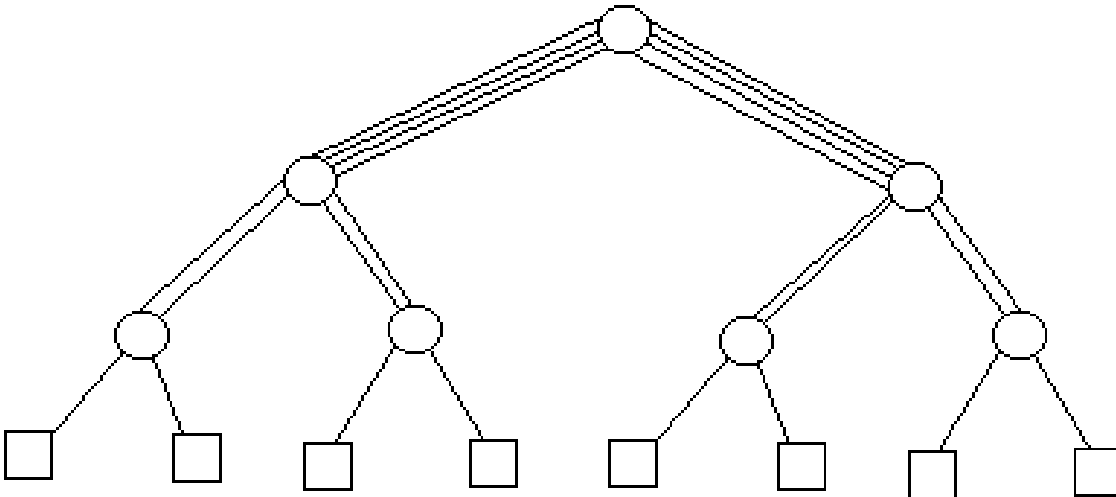
Cost, diameter, bisection width and arc connectivity

	diameter	bisection width	arc connectivity	Cost
Star	2	1	1	$p-1$
Fat Tree	$2 \log ((p+1)/2)$	1	1	$n \times 2^n$

p = number of nodes

n = number of levels

For the following tree, we have number of nodes 15 and levels 3, starting from 0.



In terms of above evaluation parameters, star has less cost and diameter however all traffic has to be passed through center node. On the other hand, fat tree manages the traffic well with distributed nodes and larger bandwidth at edges with large traffic.

Question 4: [6 + 4 marks, CLO # 2]

(i) Show the output of the following program:

```
#include <iostream>
#include <omp.h>
using namespace std;
int main() {
    int nums[10] = { 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 };
    omp_set_num_threads(3);
#pragma omp parallel for schedule(static, 3)
    for (int j = 0; j < 10; j++) {
        nums[j] *= (j+3);
        int x = omp_get_thread_num();
        cout << "At thread: " << x << " iteration: ";
        cout << j << endl;
    }

    for (int i = 0; i < 10; i++) {
        cout << nums[i] << " ";
    }
    cout << endl;

    return 0;
}
```

```
At thread: 0 iteration: 0
At thread: 0 iteration: 1
At thread: 0 iteration: 2
At thread: 1 iteration: 3
At thread: 1 iteration: 4
At thread: 1 iteration: 5
At thread: 2 iteration: 6
At thread: 2 iteration: 7
At thread: 2 iteration: 8
At thread: 0 iteration: 9
3 8 15 24 35 48 63 80 99 120
```

(ii) In general, what would you have to do if you are not allowed to use “OpenMP for construct” and you have to explicitly map the iterations to threads? Illustrate your answer with OpenMP code or pseudocode

We would have to make sure the iterations managed by different threads are mutually exclusive. We may use the thread-ID for this purpose (as a different value to initialize each threads iterations, and the number of threads as a separation or jump value between each iteration for a specific thread.

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Student : Name: **SOLUTION** Roll No. _____ Section: _____

Instruction: Attempt all questions on the question paper. Rough sheets can be used but it should not be attached. If you think some information is missing then assume it and mention it clearly.

Question 1: MCQs & True/False

[10 marks]

- i. Single Program with Multiple Data (SPMD) is a close variant of _____ in Flynn taxonomy:
 - a. SIMD
 - b. **MIMD ***
 - c. MISD
 - d. SISD

- ii. In case of _____ decomposition, divide and conquer strategy is used.
 - a. **Recursive ***
 - b. data
 - c. exploratory
 - d. Speculative

- iii. In Bus network, the number of shared connections is equal to
 - a. 1
 - b. **n ***
 - c. $\log_2 n$
 - d. n^2

- iv. The number of tasks that can be executed in parallel is called _____.
 - a. dependency
 - b. **degree of concurrency ***
 - c. granularity
 - d. All of the above

- v. Which one is true about Amdahl's law formulation: -
 - a. It does not account for communication overheads
 - b. It provides an upper-bound on the achievable speedups
 - c. If the number of processors approaches infinity, Speedup is bounded by $\frac{1}{\text{sequential fraction}}$
 - d. **All (a), (b), and (c) are correct ***
 - e. Only (b) and (c) are correct

- vi. Choose correct statement/s regarding NUMA architecture
 - a. **Access time may vary depending on the data location ***
 - b. Different processors interact with each other using message passing
 - c. **The same logical address on different processors maps to the same physical memory location ***
 - d. The same logical address on different processors maps to the different physical memory locations

- vii. **How is a network of workstations different from a cluster?**
 - a. The workstations in such a network may have different operating systems
 - b. The nodes within a network will normally be co-located (located together)
 - c. Users normally have the power to login to individual workstations
 - d. **Both (a) and (c) ***
- viii. **In distributed Systems, processors can share logical address space**
 - a. True
 - b. **False ***
- ix. **Mesh network provides better cost scalability but poor performance scalability than the bus network**
 - a. True
 - b. **False ***
- x. **In case of Non-uniform Memory access, a processor cannot directly access the memory associated with another processor**
 - a. **True ***
 - b. False

Question 2: Differentiate between

[4 marks]

- a. Exploratory and Speculative decomposition

Exploratory: Specially used to decompose the problems having underlying computation like search-space exploration.

We may use these steps:

- (1) **Partition the search space into smaller parts;**
- (2) **Search each one of these parts concurrently, until the desired solutions are found**

Speculative: Usually used in the problems where different input values or output of the previous stage causes many computationally intensive branches.

Possible steps:

- (1) **Speculate (guess) the output of previous stage**
- (2) **Start performing computations in the next stage even before the completion of the previous stage.**
- (3) **After the output of the previous stage is available, if the speculation was correct, then most of the computation for the next step would have already been done.**

- b. Static and Dynamic task mapping

Static: when the size and number of tasks are known apriori, we prefer static mapping. This mapping of tasks to processes is done before program execution.

Dynamic: when the size and number of tasks are not known apriori, we prefer dynamic mapping. This mapping of tasks to processes is done during program execution.

Question 3: Assume a sequential program S has an execution time of 200 seconds. Further, assume that S_p is a parallel variant of S . After an experimental evaluation over different number of processors, the following running times were achieved: - [4+4 marks]

P	2	4	6	8
Execution Time (seconds)	107	61.92	48.31	42.46
Speedup	1.87	3.23	4.14	4.71
Karp-Flatt Metric	0.07	0.08	0.09	0.10

- a) Calculate Speedups for each of the experimental configurations in the space provided below and then write your answers in the table above.

$$200 / 107 = 1.869158 = 1.87$$

$$200 / 61.92 = 3.229974 = 3.23$$

$$200 / 48.31 = 4.1399296 = 4.14$$

$$200 / 42.46 = 4.710315 = 4.71$$

- b) Calculate the Karp-Flatt metric values in the space provided below and then write your answers in the table above. You also have to Interpret the results of Karp-Flatt metric and write your opinion below

$$((1/1.87) - (1/2)) / (1 - 1/2) = 0.06951871657 = 0.07$$

$$((1/3.23) - (1/4)) / (1 - 1/4) = 0.07946336429 = 0.08$$

$$((1/4.14) - (1/6)) / (1 - 1/6) = 0.08985507246 = 0.09$$

$$((1/4.71) - (1/8)) / (1 - 1/8) = 0.09978768577 = 0.1$$

As 'e' is steadily increasing with p, it suggests that parallelization overhead is also contributing to poor speedup. Therefore, we need to reduce this overhead to improve speedups.

Question 4: For this question, you have a choice to attempt any one of the following questions: [8 marks]

a) Derive generalized expressions for arc-connectivity, diameter, bisection-width, and link cost of a 2D mesh (without wraparound) having M rows and N columns.

OR

b) Write a multithreaded program either using Open MP or Pthreads to find the maximum value in a matrix of size n x m. (Minimum number of threads should be 2 and maximum could be n)

(a)

Arc connectivity = 2 (each arc is connected to two nodes)

Diameter = (Row distance + Column distance) = (N-1) + (M-1) = M + N - 2

Bisection width = Minimum (M, N) (i.e., the min. number of links that must be removed to partition in equal halves)

Cost = (Links in a Row)(Total Rows) + (Links in a column)(Total Columns)

= (N-1)(M) + (M-1)(N) = MN - M + MN - N = 2MN - (M + N)

(b) **One Possible solution:**

```
pthread_mutex_lock maxlock; // global variables
int max = -10000;
int** matrix;

void function(void* arg) {
    int row = * (int*) arg;
    int localMax = -1000;
    for(int i=0; i<M; i++) {
        if(matrix[row][i] > localMax) localMax = matrix[row][i];
    }
    mutex_lock(maxLock);
    if(localMax > max) max = localMax;
    mutex_unlock(maxLock);
    pthread_exit(NULL);
}

int main() {
    int N, M; // values from user
    matrix = new int*[N];
    for(int i=0; i<N; i++) matrix[i] = new int[M];
    // matrix is filled by user
    pthread_t* threads;
    threads = new pthread_t[N];

    for(int i=0; i<N; i++) pthread_create(&threads[i], NULL, function, (void *) i);
    for(int i=0; i<N; i++) pthread_join(&threads[i], NULL);

    cout << "Max " << max << endl;
    delete [] threads;
    for(int i=0; i<N; i++) delete [] matrix[i];
    delete matrix;
    return 0;
}
```